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OPTICS
AND LASER PHYSICS

Lineshape-Asymmetry-Caused Shift in Atomic Interferometers

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We investigate the shift caused by asymmetry of spectroscopic lineshape in atomic interferometers, which has not previously been discussed in the scientific literature. This asymmetry arises because laser field is frequency-chirped not only during the free-evolution intervals of atoms, but also during the Ramsey pulses. As a result, the effective detuning from the working atomic transition during the pulses also depends on the chirping rate, which, in turn, leads to the lineshape-asymmetry-caused shift (LACS). It is shown that this shift has an inverse cubic dependence of $\propto 1/T^3$ on the duration of the interval between the Ramsey pulses T, which markedly contrasts with the $\propto 1/T^2$ dependence typical in atomic interferometry. Therefore, the metrological importance of this shift substantially increases for compact atomic interferometers with a short baseline. For example, for interferometers-gravimeters using two-photon transitions in rubidium atoms, at $T \sim 1$ ms we estimate the LACS shift and its variations at the level of 0.1–1 mGal, while for $T \sim 100$ μs this can reach a value of 0.1–1 Gal.

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Atomic interferometry is the basis of modern quantum sensors for various purposes. For example, quantum absolute gravimeters based on the interference of ultracold atoms were first implemented in 1991 [1] and in the next demonstrated outstanding results in the field of high-precision measurements of the free-fall acceleration [2–5]. Their high sensitivity, long-term stability, and lack of need for calibration make them an indispensable tool for a wide range of fundamental [6–10] and practical applications [11, 12].

Current trends in the development of atomic interferometers are aimed at miniaturization and increased mobility, which inevitably leads to a reduction of the transit time of atoms in the interferometer. For example, in high-sensitive transportable quantum gravimeters, the interferometric cycle time can reach several hundred milliseconds [3]. This, in turn, significantly limits the compactness, response time, and dynamic range of the sensor, and increases its sensitivity to sources of phase noise and systematic shifts such as low-frequency vibration, residual magnetic field gradients, spatial inhomogeneity of the intensity and phase of light fields, etc.

In this context, short-baseline atomic interferometers (i.e., with small atomic flight time) represent a promising alternative. They exhibit significantly lower sensitivity to external disturbances, high sampling rates (tens of hertz), and great potential for miniaturization. Moreover, as shown in [13, 14], even with interferometric atomic sampling sequences on the order of milliseconds, it is possible to achieve metrological characteristics that meet the requirements of most practical applications. This is facilitated by the development of new approaches to increasing the phase stability and contrast of interference signals [15–17], as well as effective methods for analyzing, correcting, and processing measurements [18, 19]. Thus, short-baseline interferometers represent a promising direction for the development of high-precision compact quantum sensors.

In this paper, we present a theoretical study of the systematic shift caused by the asymmetry of the signal

lineshape in atomic interferometers, which has not previously been discussed in the scientific literature. In particulary, we demonstrate that for short-baseline interferometer-gravimeters, this shift can be dominant, significantly limiting the accuracy and stability of compact devices.

As a specific scheme of an atomic interferometer, we will consider a quantum gravimeter using an atomic transition $|g\rangle \leftrightarrow |e\rangle$ with an unperturbed frequency ω_0, where $|g\rangle$ and $|e\rangle$ are the ground and excited states, respectively, which correspond to the following column vectors:

$$|g\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad |e\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{1}$$

In this two-level atom, transitions between states $|g\rangle$ and $|e\rangle$ during Ramsey pulses are induced under the action of a resonant interaction operator of the plane running wave type with the wave vector $\mathbf{k}_{\text{eff}}$

$$\hat{V}(t,\mathbf{r}) = -\frac{\hbar\Omega_0}{2} e^{-i[\omega t+\alpha t^2/2-(\mathbf{r}\cdot\mathbf{k}_{\text{eff}})+\phi]} |e\rangle\langle g| + \text{H.c.}, \tag{2}$$

where ω is the carrier frequency, which can vary near the transition frequency ω_0, α is the speed of the additional frequency chirp, ϕ is some fixed phase, and $\Omega_0 > 0$ is the real Rabi frequency of the induced transitions. As is known, in such an interferometric scheme, the quantity $(\mathbf{g}\cdot\mathbf{k}_{\text{eff}})$ can be measured, where $\mathbf{g}$ is the free fall acceleration due to gravity.

Note that the interaction type (2) occurs both for one-photon transition under the action of a running light wave in the case of the optical transition $|g\rangle \leftrightarrow |e\rangle$, and for two-photon transition in the bichromatic field of two counterpropagating running light waves, when the difference in the optical frequencies of these waves $(\omega_1 - \omega_2)$ varies near the microwave frequency ω_0 in the case of the transition $|g\rangle \leftrightarrow |e\rangle$ between hyperfine levels of an atom in the ground state (for example, in alkali metal atoms). However, this does not matter for our work.

As an example, we will consider a standard Mach–Zehnder type atomic interferometer-gravimeter based on a sequence of three Ramsey pulses ($\pi/2$–π–$\pi/2$ sequence), shown in Fig. 1. The general theory of atomic interferometry is described in [20, 21].

In the reference frame of a moving atom, the operator (2) can be represented as

$$\hat{V}^{(\text{atom})} = -\frac{\hbar\Omega_0}{2} e^{-i\varphi(t)} |e\rangle\langle g| + \text{H.c.}, \tag{3}$$

with a time-dependent phase, $\varphi(t)$. Then, for sufficiently short durations of Ramsey pulses, the evolution

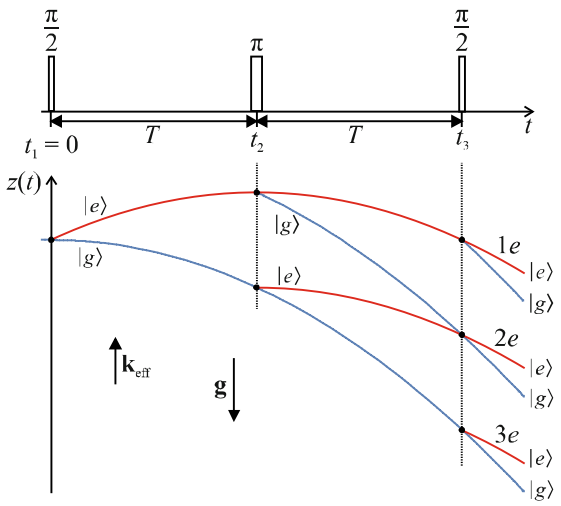

Fig. 1. (Color online) Standard scheme of a Mach–Zehnder type atomic interferometer-gravimeter.

operator of an atom under the action of the jth Ramsey pulse can be written as

$$\hat{R}^{(j)} = \begin{pmatrix} A_j & iB_j e^{-i\varphi(t_j)} \\ iB_j e^{i\varphi(t_j)} & A_j^* \end{pmatrix}, \tag{4}$$

where t_j is the point in time of the jth Ramsey pulse (see Fig. 1). The matrix elements in (4) are defined as follows

$$A_j = \cos\left(\frac{\Omega^{(j)}\tau_j}{2}\right) + i\frac{\delta_j}{\Omega^{(j)}}\sin\left(\frac{\Omega^{(j)}\tau_j}{2}\right),$$
$$B_j = \frac{\Omega_0}{\Omega^{(j)}}\sin\left(\frac{\Omega^{(j)}\tau_j}{2}\right), \tag{5}$$
$$\Omega^{(j)} = \sqrt{\Omega_0^2 + \delta_j^2},$$

where τ_j is the duration of the jth pulse, δ_j is the effective detuning from the resonance during the jth pulse from the point of view of the moving atom, $\Omega^{(j)}$ is the generalized Rabi frequency during the jth pulse. Note that B_j is a real number.

Let us consider the usual scheme of interrogating atoms in the case of spatially uniform acceleration of gravity $\mathbf{g}$. Let at time $t = 0$ (i.e., at the beginning of the first Ramsey pulse, see Fig. 1) the atoms are in the lower state $|g\rangle$ and have an initial velocity $\mathbf{v}$. Then, as the spectroscopic signal after the action of the Ramsey sequence in Fig. 1, we will consider the total population N_e of atoms in the upper state $|e\rangle$ on three different trajectories: 1e, 2e and 3e (see the red solid lines in Fig. 1). We assume that after the action of three Ramsey pulses all three trajectories have diverged by dis-

tances significantly greater than the de Broglie wavelength. In this case, we have the following expression

$$N_e = n_{1e} + n_{2e} + n_{3e} = |K_{1e}|^2 + |K_{2e}|^2 + |K_{3e}|^2. \quad (6)$$

Using the notation from (4), we can write

$$K_{1e} = B_1 A_2 A_3,$$
$$K_{2e} = B_1 B_2 B_3 - A_1^* B_2 A_3 e^{i\tilde{\alpha} T^2}, \quad (7)$$
$$K_{3e} = A_1^* A_2^* B_3,$$

where T is the time between pulses (see Fig. 1), and $\tilde{\alpha}$ is the frequency chirp speed centered on the measured value $(\mathbf{g} \cdot \mathbf{k}_{\text{eff}})$

$$\tilde{\alpha} = \alpha - (\mathbf{g} \cdot \mathbf{k}_{\text{eff}}). \quad (8)$$

In the standard interrogation scheme Fig. 1, we have the following expressions for the duration and detuning of each pulse

$$\tau_1 = \tau_3 = \tau, \quad \tau_2 = 2\tau,$$
$$\delta_1 = \Delta_{\text{D}}, \quad \delta_2 = \tilde{\alpha} T + \Delta_{\text{D}}, \quad \delta_3 = 2\tilde{\alpha} T + \Delta_{\text{D}}, \quad (9)$$

where Δ_{D} is the general residual contribution to the detuning from resonance, which, taking into account the recoil effect for atoms with mass M, has the following form

$$\Delta_{\text{D}} = \omega - \omega_0 - \frac{\hbar |\mathbf{k}_{\text{eff}}|^2}{2M} - (\mathbf{v} \cdot \mathbf{k}_{\text{eff}}) - \bar{\delta}_{\text{ac}}, \quad (10)$$

where $(\mathbf{v} \cdot \mathbf{k}_{\text{eff}})$ is the Doppler shift during the action of the first Ramsey pulse at $t = 0$ (see Fig. 1). In (10) we also take into account the ac Stark shift $\bar{\delta}_{\text{ac}}$ of the atomic transition frequency caused by the probe light field, which we assume to be the same for all three pulses.

Let us now investigate the functional dependence of the signal $N_e(\tilde{\alpha})$ on the variable parameter $\tilde{\alpha}$

$$N_e(\tilde{\alpha}) = |K_{1e}(\tilde{\alpha})|^2 + |K_{2e}(\tilde{\alpha})|^2 + |K_{3e}(\tilde{\alpha})|^2. \quad (11)$$

In general, the dependence $N_e(\tilde{\alpha})$ on the variable $\tilde{\alpha}$ is a sequence of narrow Ramsey resonances with a width of about π/T^2 and a smooth envelope (the width is of the order of $\sim 2\Omega_0/T$), which is described by the interference contribution $|K_{2e}(\tilde{\alpha})|^2$ in (11). In addition, there is also a wide smooth dependence (the width is of the order of $\sim 2\Omega_0/T$), which is formed by two other terms $|K_{1e}(\tilde{\alpha})|^2$ and $|K_{3e}(\tilde{\alpha})|^2$. The typical form of the corresponding dependences is shown in Fig. 2.

In this case, the gravimetric measurement of the value $(\mathbf{g} \cdot \mathbf{k}_{\text{eff}})$ corresponds to the position $\tilde{\alpha}_{\text{ext}}$ of the central Ramsey resonance peak near zero (see Fig. 3). The $\tilde{\alpha}_{\text{ext}}$ value itself determines the shift of the gravimetric measurement. Based on the expressions (5), (7) and (9), it is easy to show that in the case of $\Delta_{\text{D}} = 0$ the following relations are satisfied: $K_{1e}(-\tilde{\alpha}) = K_{1e}^*(\tilde{\alpha})$, $K_{2e}(-\tilde{\alpha}) = K_{2e}^*(\tilde{\alpha})$ and $K_{3e}(-\tilde{\alpha}) = K_{3e}^*(\tilde{\alpha})$ [because of $\delta_j(-\tilde{\alpha}) = -\delta_j(\tilde{\alpha})$]. However, for $\Delta_{\text{D}} \neq 0$ these relations are violated: $K_{1e}(-\tilde{\alpha}) \neq K_{1e}^*(\tilde{\alpha})$, $K_{2e}(-\tilde{\alpha}) \neq K_{2e}^*(\tilde{\alpha})$ and $K_{3e}(-\tilde{\alpha}) \neq K_{3e}^*(\tilde{\alpha})$ [because of $\delta_j(-\tilde{\alpha}) \neq -\delta_j(\tilde{\alpha})$]. As a result, we have the following relations

$$N_e(-\tilde{\alpha}) = N_e(\tilde{\alpha}) \quad \text{for} \quad \Delta_{\text{D}} = 0,$$
$$N_e(-\tilde{\alpha}) \neq N_e(\tilde{\alpha}) \quad \text{for} \quad \Delta_{\text{D}} \neq 0. \quad (12)$$

Thus, for $\Delta_{\text{D}} = 0$ the signal $N_e(\tilde{\alpha})$ is an even (symmetric) function on $\tilde{\alpha}$, and, therefore, the position of the local extremum at the central Ramsey resonance is

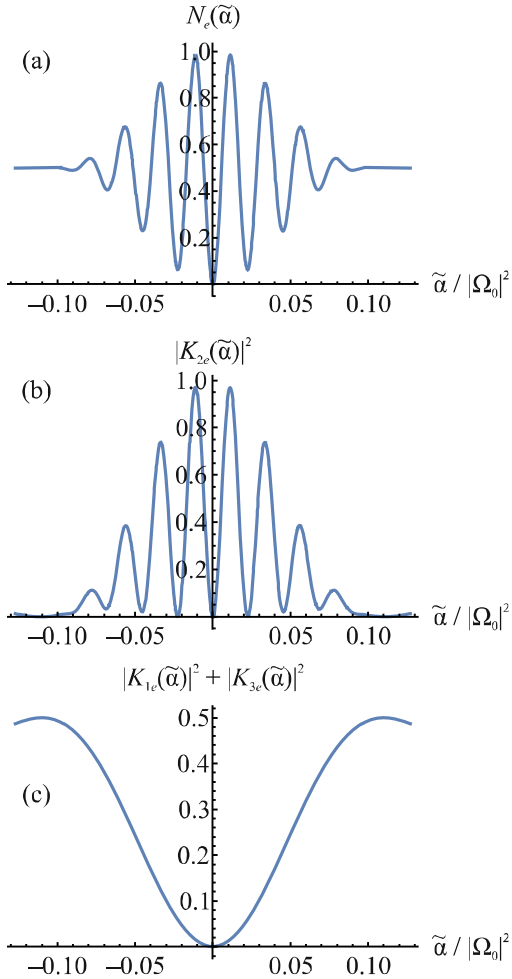

Fig. 2. (Color online) Typical forms of dependencies: (a) total signal $N_e(\tilde{\alpha})$; (b) interference contribution $|K_{2e}(\tilde{\alpha})|^2$; (c) substrate $(|K_{1e}(\tilde{\alpha})|^2 + |K_{3e}(\tilde{\alpha})|^2)$. Calculation parameters: $\tau\Omega_0 = \pi/2$, $T = 10\tau$, $\Delta_{\text{D}} = 0$.

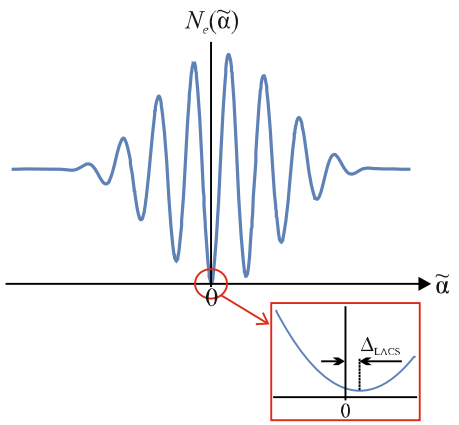

Fig. 3. (Color online) Demonstration of the shift in the position of the extremum at the central Ramsey resonance in the dependence $N_e(\tilde{\alpha})$ under $\Delta_D \neq 0$.

unshifted, i.e. $\tilde{\alpha}_{ext} = 0$. However, for $\Delta_D \neq 0$, the signal $N_e(\tilde{\alpha})$ is no longer a symmetric function on $\tilde{\alpha}$, which means, in our case, a shift in the position of the central Ramsey resonance peak: $\tilde{\alpha}_{ext} = \Delta_{LACS} \neq 0$ (see Fig. 3). This can be characterized as a shift caused by the lineshape asymmetry (lineshape-asymmetry-caused shift, LACS). Calculations show that in the case of $\Omega_0\tau = \pi/2$ and $|\Delta_D| < 0.5\Omega_0$, the following linear dependence on Δ_D is well satisfied:

$$\Delta_{LACS} \approx \xi_1(T/\tau)\frac{\tau}{T^3}\frac{\Delta_D}{\Omega_0}, \tag{13}$$

where the dimensionless function $\xi_1(T/\tau) < 0$ depends rather slowly on the ratio T/τ in the case of $T \gg \tau$ (see Fig. 4). Thus, the dependence of the detunings $\delta_j(\tilde{\alpha})$ on the parameter $\tilde{\alpha}$ [see expressions in (9)] and the presence of general residual contribution $\Delta_D \neq 0$ to the detunings from the resonance during the action of the Ramsez pulses leads to a shift $\Delta_{LACS} \neq 0$ when measuring the quantity $(\mathbf{g} \cdot \mathbf{k}_{eff})$. It should be noted that this shift is not a phase shift in the standard meaning, but is the result of asymmetry in both the envelope of the interference term $|K_{2e}(\tilde{\alpha})|^2$ and the substrate $\left(|K_{1e}(\tilde{\alpha})|^2 + |K_{3e}(\tilde{\alpha})|^2\right)$.

In the case of a non-monovelocity atomic ensemble, it is necessary to consider the signal averaged over the velocity distribution $f(\mathbf{v})$:

$$\langle N_e(\tilde{\alpha})\rangle_{\mathbf{v}} = \int f(\mathbf{v})N_e(\tilde{\alpha})d\mathbf{v}^3, \tag{14}$$

and the LACS value will undergo significant changes relative to the expression (13). For example, consider

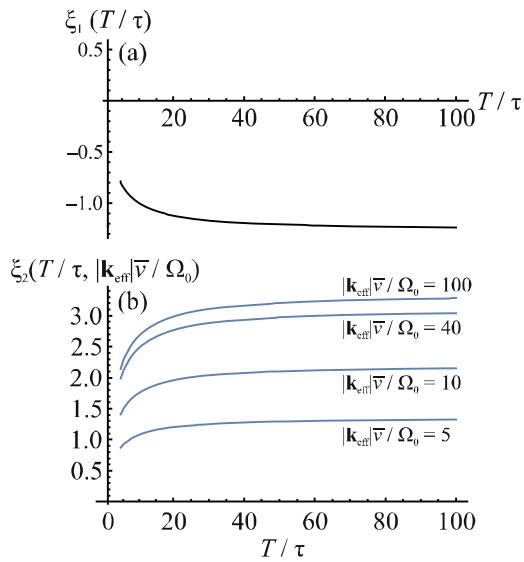

Fig. 4. (Color online) (a) Dependence of the dimensionless function $\xi_1(T/\tau)$ in the expression (13). (b) Dependence of the dimensionless function $\xi_2(T/\tau, |\mathbf{k}_{eff}|\bar{v}/\Omega_0)$ on T/τ in the expression (16) for several fixed values $|\mathbf{k}_{eff}|\bar{v}/\Omega_0 = 5, 10, 40, 100$ (see from bottom to top).

an atomic ensemble with a Maxwellian velocity distribution

$$f_M(\mathbf{v}) = \frac{1}{\pi^{3/2}\bar{v}^3}e^{-(\mathbf{v}-\mathbf{v}_0)^2/\bar{v}^2}, \tag{15}$$

where $\bar{v}$ is the width parameter of the velocity distribution relative to the average velocity of the atomic beam $\mathbf{v}_0$. In the general case, there is no simple expression for LACS. However, for $\Omega_0\tau = \pi/2$ and $T \gg \tau$, in the case where the Doppler width $|\mathbf{k}_{eff}|\bar{v}$ significantly exceeds the Rabi frequency, $|\mathbf{k}_{eff}|\bar{v} > 5\Omega_0$, the following expression works quite well

$$\Delta_{LACS}^{\langle\mathbf{v}\rangle} \approx \xi_2(T/\tau, |\mathbf{k}_{eff}|\bar{v}/\Omega_0)\frac{\tau}{T^3}\frac{\Omega_0\langle\Delta_D\rangle_{\mathbf{v}}}{(|\mathbf{k}_{eff}|\bar{v})^2}, \tag{16}$$

where $\langle\Delta_D\rangle_{\mathbf{v}}$ is the residual Doppler detuning for the average atomic velocity $\mathbf{v}_0$:

$$\langle\Delta_D\rangle_{\mathbf{v}} = \omega - \omega_0 - \frac{\hbar|\mathbf{k}_{eff}|^2}{2M} - (\mathbf{v}_0 \cdot \mathbf{k}_{eff}) - \bar{\delta}_{ac}, \tag{17}$$

and $\xi_2(T/\tau, |\mathbf{k}_{eff}|\bar{v}/\Omega_0)$ is a dimensionless function that depends rather slowly on two dimensionless parameters T/τ and $|\mathbf{k}_{eff}|\bar{v}/\Omega_0$ (see Fig. 4b). Formula (16) is well satisfied for $|\langle\Delta_D\rangle_{\mathbf{v}}| < 0.5|\mathbf{k}_{eff}|\bar{v}$. Comparing the signs of the dependences in Figs. 4a and 4b, it is interesting to note that the shift (16) is opposite in sign to the shift for a mono-velocity atomic ensemble (13).

Let us estimate the metrological significance of the LACS using the example of atomic gravimeter-interferometer. Obviously, due to the inverse cubic dependence ($\propto 1/T^3$), the role of this shift in the error budget will increase for compact interferometers with a short baseline. Consider, for example, the following parameters of the Ramsey atomic interrogation scheme: $T = 2.5$ ms, $\tau = 25$ μs, and $\Omega_0 = 2\pi\times 10$ kHz (i.e., $\Omega_0\tau = \pi/2$). With regard to the velocity distribution of the atomic beam, we use the following values: $|\mathbf{v}_0| \sim 50$ cm/s и $\bar{v} \sim 5$ cm/s, which for two-photon transition through the D lines in rubidium atoms gives estimates of $|(\mathbf{v}_0 \cdot \mathbf{k}_{\text{eff}})| \sim 1$ MHz and $|\mathbf{k}_{\text{eff}}|\bar{v} \sim$ 100 kHz. In this case, taking into account the possibility of a one-percent variation in the value of $|\mathbf{v}_0|$ during gravimetric measurements, we can safely assume an estimate of $|\langle\Delta_{\text{D}}\rangle_{\mathbf{v}}| \gtrsim 10$ kHz both for the value of $|\langle\Delta_{\text{D}}\rangle_{\mathbf{v}}|$ itself and for its variations. Substituting these values into the formula (16), we obtain an estimate of the LACS value and its variations at the level of $\left|\Delta_{\text{LACS}}^{\langle\mathbf{v}\rangle}\right| \gtrsim 20$ s^{-2}, which, when applied to gravimetry, corresponds to $\gtrsim$0.1 mGal. Thus, on the millisecond scale of free evolution times in atomic interferometer-gravimeters, the LACS shift can lead to measurement errors and fluctuations at the level of 0.1–1 mGal. Accordingly, for a free evolution time $T \sim 100$ μs, this estimate reaches values of 0.1–1 Gal. Obviously, shifts of this magnitude can significantly reduce the metrological characteristics of short-baseline atomic interferometers. Moreover, in the case of a mono-velocity atomic beam, this shift can reach even greater values. We also note that although for interferometer-gravimeters with a free evolution time at the level of tens of milliseconds the LACS shift is not so critically large, it is nevertheless not negligibly small (it can reach the order of 1–10 μGal) and it should also be taken into account in the measurement error budget.

For the sake of general discussion, it should be noted that the most compact quantum gravimeters owe their efficiency to a high interrogation rate of atoms (tens of hertz) and the recapturing of cold atoms in the vapor cell into a magneto-optical trap at the end of the experimental cycle. This, in turn, sets the total free-fall time of atoms at 40–50 ms (8–12 mm), determined by the size of the laser beam overlap region, the recapture efficiency, the required response time, etc. During the free-fall time of atoms, three stages occur: preparation of the initial atomic wave packet, atomic interference, and detection. The duration of the atom preparation stage depends on the preparation method, but averages several tens of milliseconds. The duration of the detection stage after the Ramsey sequence is typically several milliseconds [2]. The remaining time (approximately 20 ms or less) is accounted for by the atomic interference stage itself. An example of a compact gravimeter was demonstrated in [13], where the T value did not exceed 7 ms, and the sensitivity reached from 570 to 36.7 mGal Hz$^{-1/2}$ at interrogation rates from 50 to 330 Hz, respectively. It should be noted that gravimeters with a short time T have a high response time, increased stability to many systematic shifts (e.g., magnetic field gradient, spatial non-uniformity of intensity and wavefront curvature, etc.) and to vibrations, making it possible to use quantum gravimeters in highly dynamic conditions and on moving platforms. All of the above, taking into account the general trend of device miniaturization, demonstrates the relevance of our results for atomic gravimeters, in which the interval between Ramsey pulses does not exceed 10–20 ms, and especially at T of the order of milliseconds or less. Moreover, in addition to quantum gravimeters, one can also consider other devices (for example, atomic accelerometers and gyroscopes) that use chirping of the probe laser frequency.

In conclusion, we theoretically investigated the shift in atomic interferometers, which has not previously been discussed in the scientific literature. This shift is caused by the lineshape asymmetry of the spectroscopic signal (LACS) when using the frequency-chirped probe field method and has an inverse cubic dependence of $\propto 1/T^3$ on the duration of the interval between the Ramsey pulses T, which contrasts markedly with the typical dependence for atomic interferometry $\propto 1/T^2$. Thus, its influence increases significantly for atomic interferometers with a short baseline. Note that this shift is not a phase shift in the standard meaning, but is the result of asymmetry in both the envelope of the interference term and the substrate. Using the example of a Mach–Zehnder type interferometer-gravimeter, it was shown that even starting with T of tens milliseconds, the LACS shift can have metrological significance, and at T of milliseconds and below, it can become dominant, substantially limiting the accuracy and stability of small-sized devices. Therefore, its experimental detection, detailed study, and development of suppression methods are relevant tasks for atomic interferometry and the creation of precision compact and mobile quantum sensors for various purposes (gravimeters, accelerometers, and gyroscopes).

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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